** Useful data: 760 mmHg =1 atm = 760 torr =101.3 Kpa, R=0.0821 atm.L.mol^{-1}.K^{-1}
Awt: C=12 /Na =23 /Ca = 40/H = 1 / S = 32 / O = 16 / N= 14 / Cl = 35.5 g/mol.

Mm: $CaSO_4$= 136, Na_2SO_3=126 g/mol.

Q1: Choose the correct answer for the following questions and transfer it to the table below:

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	A	B	C	C	B	B	D	B	A

1) The incorrect statement of the following is:
A- You can heat all types of liquid in blue flame directly.
B- Burette is one of the most precises volumetric glass wares.
C- Acid and Bases are example of corrosive materials.
D- When some chemicals spilled on your hand washing with directly with tap water.

2) You can infer that the substance is toxic due to the pictogram:
A-  B-  C-  D- 

3) Which one of these chemical salts will shows clear solution with water?
A- AgCl B- $NaNO_3$ C- PbI_2 D- $CaCO_3$

4) During Density experiment, the data in the table beside are obtained, calculate the final volume of water in the cylinder used in this measurement:
A- 18.5 mL B- 14.9 mL
C- 17.2 mL D- 17.6 mL

The density of solid	4.63 g/cm^3
Initial volume of water	13.7 ml
Mass of solid	16.2 g

5) Which one of these compounds give you the lowest boiling point:
A- CH_3CH_2OH B- $CH_3(CH_2)_4CH_2OH$ C- C_2H_2 D- $CH_3(CH_2)_4CH_3$

6) Which one of the following statements is correct concerning Boiling point experiment:
A- The experiment indicates that the atmospheric pressure is more than the vapor pressure of liquid.
B- The boiling point of a liquid can measured when air bubbles are come out from liquid.
C- Boiling of liquid takes place at all temperature.
D- Boiling point is physical property is used to indicate the purity of liquids.

7) Calculate the mass percent of water in $CaSO_4.3H_2O$, if the mass of anhydrous = 5.7g.
A- 20.8% B- 32.4% C- 28.4% D- 40.0%

8) A hydrate of Na_2SO_3 has 30% water by mass. It is.....
A- $Na_2SO_3 \cdot 4H_2O$
B- $Na_2SO_3 \cdot 6H_2O$
C- $Na_2SO_3 \cdot 7H_2O$
D- $Na_2SO_3 \cdot 3H_2O$

9) Which one of these liquids you can find its molecular weight by applying dumas method?
A- Toluene (B.P= 121 °C)
B- Butanal (B.P= 76 °C)
C- Aniline (B.P= 184 °C)
D- Benzonitrile (B.P= 190 °C)

10) According to the data in the table below, calculate the molecular weight of volatile liquid:

A- 43.5 g/mol
B- 47.5 g/mol
C- 51.5 g/mol
D- 58.5 g/mol

Mass of empty flask	65.35 g
Mass of flask + vapor	65.74 g
Temperature of vapor	384 K
External pressure	730 mmHg
Mass of flask filled with water	327 g
Density of water.	0.97 g/ml

Q2: True or False Questions:

1. Fume hood is a suitable equipment that used to getting rid of toxic gases. (✓)
2. The solubility of gas decrease when temperature decrease. (✗)
3. Hydrates is the name which given to a crystal salt that contains water of crystallization. (✓)
4. Ethanol can be heated to measured its boiling point directly on blue flame. (✗)
5. A suitable glassware used to determine the volume of the gas in molecular weight of volatile liquid experiment was a graduated cylinder. (✓)

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Q3: Fill in the blanks:

1. The uncertainty of the analytical balance is..0.0001.....
2. In density experiment, if a partial of a solid dissolved and the volume of water in cylinder is increased, the calculated density will be..decrease.....
3. If there is some glass in the liquid, then the measured boiling point will be..not effect.....
4. Using yellow flame cause the calculated the percentage of water to be..lower.....
5. In determination molecular weight of volatile liquid, if a volume written on the flask was used instead of the real one, then the calculated molar mass will be..different.....

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GOOD LUCK